

BISHWASH KHANAL

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EDUCATION

Master of Science in Artificial Intelligence

Aug 2024 - Present

University of Jyväskylä, Finland

Relevant Courses: Deep Learning for Cognitive Computing, Semantic Web, SOA and Cloud Computing, Computer Vision and Image Analysis, Natural Language Processing

Bachelor of Science in Electrical and Computer Engineering

Sep 2017 - Aug 2020

Jacobs University Bremen, Germany

Minor: Intelligent Mobile Systems, Grade: 1.73 (German grading system; 1: Very Good, 5: Insufficient)

Relevant Courses: Machine Learning, Computer Vision, Artificial Intelligence, Signal Processing, Robotics

Thesis: Non-Linearity in Wireless Communications and Deep Learning [🔗](#)

PUBLICATIONS

- B. Khanal, M. Om, S. Rijal, V. Ojha (2024). Alignment of a 360° image with posed color images for locally accurate texturing of 3D mesh. *Frontiers in Computer Science*, Vol. 6, ISSN: 2624-9898. [doi:10.3389/fcomp.2024.1388174](https://doi.org/10.3389/fcomp.2024.1388174)
- B. Khanal, S. Rijal, M. Awale, V. Ojha (2024). Structure-preserving Planar Simplification for Indoor Environments. *arXiv preprint arXiv:2408.06814*. [doi:10.48550/arXiv.2408.06814](https://doi.org/10.48550/arXiv.2408.06814)
- B. Khanal, J. M. Capone (2024). Evaluating the Impact of Compression Techniques on Task-Specific Performance of Large Language Models. *arXiv preprint arXiv:2409.11233*. [doi:10.48550/arXiv.2409.11233](https://doi.org/10.48550/arXiv.2409.11233)
- T. Hasanli, S. Siddeeq, B. Khanal, P. Kotilainen, T. Mikkonen, P. Abrahamsson (2026). TDD Governance for Multi-Agent Code Generation via Prompt Engineering. *arXiv preprint arXiv:2604.26615*. [doi:10.48550/arXiv.2604.26615](https://doi.org/10.48550/arXiv.2604.26615)

SKILLS

Programming Languages: Python, C, C++, MATLAB, Java

Machine Learning: TensorFlow, PyTorch, Hugging Face, LLMs, Transformers, GANs, CNNs

Computer Vision: OpenCV, Open3D, TensorRT, ONNX

Tools & Technologies: Docker, Kubernetes, Git, FastAPI, LangChain, $\LaTeX$

Databases: MySQL, PostgreSQL, ChromaDB

WORK EXPERIENCE

Research Assistant - ANSE Project

Jan 2026 - Present

University of Jyväskylä (part-time, hybrid)

Jyväskylä, Finland

- Design and iterate on TDD orchestration workflows for AI-native software engineering
- Build and evaluate multi-agent systems for coding tasks, including automated test-driven development pipelines
- Research challenges and failure modes in TDD-based systems and propose practical mitigations
- **Technologies:** Python, Pytest, LangGraph, OpenAI API, Docker

Artificial Intelligence Engineer

Mar 2024 - Aug 2025

OptiML Org (part-time, remote)

California, USA

- Developed custom pipelines for fine-tuning and compressing open-source LLMs (Llama, Qwen, Mistral, Gemma, Phi, Deepseek), evaluating performance on domain-specific tasks
- Researched and evaluated state-of-the-art LLM compression techniques integrating model compression into fine-tuning workflows
- Founding team member focused on empowering efficient deployment of large models at scale
- **Technologies:** PyTorch, Transformers, PEFT, BitsAndBytes, Accelerate, WandB

Computer Vision Engineer

Mar 2021 - Aug 2024

E.K. Solutions Pvt. Ltd. (full-time, on-site)

Lalitpur, Nepal

- Quantized and deployed face detection, recognition, and swapping models (YOLOv8, ARCFace, Inswapper, StyleGAN2) on Jetson AGX Orin using INT8 quantization with PyTorch and TensorRT for real-time edge performance
- Developed RAG-based chatbot integrating GPT-4 for project management, database analysis, and bulk CV screening using LangChain, ChromaDB, and FastAPI
- Generated planar simplified 3D meshes for virtual tours from iPad Pro LiDAR captures, implementing custom algorithms for planar simplification, ICP registration, TSDF volumetric integration, and MVS texturing
- Applied RandLA-Net for semantic segmentation and developed full-stack pipeline using Python, C++, Swift, and Three.js
- **Technologies:** PyTorch, TensorRT, ONNX, OpenCV, Open3D, LangChain, FastAPI, Celery, NumPy

Data Annotation Specialist
Scale AI (freelancing, remote)

Jun 2023 - Oct 2023
California, USA

- Contributed to training Generative AI models by solving competitive programming problems, ranking AI-generated responses, and assessing factual accuracy of AI-produced text

Computer Vision Engineer
GoThru Media Inc. (part-time, remote)

Apr 2023
Ontario, Canada

- Automated pose estimation between uncalibrated spherical panoramas, eliminating manual alignment processes

RESEARCH PROJECTS

- **Intelligent Legal Document Analysis Platform (Lawgic)** Aug 2025 - Jan 2026
Developed a full-stack legal document analysis platform using FastAPI and React, integrating an external OCR microservice for processing PDF uploads. Developed a conversational AI interface with features for multi-lingual document translation, semantic annotations, and automated legal analysis.
- **Nepali OCR using Vision Transformers** Sep 2023
Developed vision-based transformer model for Nepali text detection, trained RoBERTa-based tokenizer and encoder on Nepali datasets
- **Face Generator using Generative Models** Mar 2023 - Jun 2023
Implemented autoencoders, VAE, DCGAN, and WGAN to generate synthetic facial images using CelebA dataset with comprehensive model evaluation
- **Google ASL Fingerspelling Recognition (Kaggle)** May 2023 - Aug 2023
Built TensorFlow Lite model using MediaPipe Holistic for American Sign Language translation, ranking 371/1315 teams

CERTIFICATIONS

- Deep Learning Specialization, Coursera [🔗](#)
- Natural Language Processing Specialization, Coursera [🔗](#)
- Generative Adversarial Networks Specialization, Coursera [🔗](#)
- Machine Learning Operations (MLOps) Specialization, Coursera [🔗](#)